```
b3ast@UltraBot-VM:~$ # SS1. System Info & Internet Check
b3ast@UltraBot-VM:~$ uname -a
Linux UltraBot-VM 6.17.0-20-generic #20~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Mar 19 01:28:37 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
b3ast@UltraBot-VM:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
b3ast@UltraBot-VM:~$ ping -c 4 google.com
PING google.com (142.250.76.206) 56(84) bytes of data.
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=1 ttl=128 time=27.5 ms
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=2 ttl=128 time=25.7 ms
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=3 ttl=128 time=29.6 ms
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=4 ttl=128 time=52.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3076ms
rtt min/avg/max/mdev = 25.662/33.855/52.659/10.945 ms
b3ast@UltraBot-VM:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=19.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=12.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=9.99 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=24.1 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3201ms
rtt min/avg/max/mdev = 9.993/16.368/24.086/5.671 ms
```

```
b3ast@UltraBot-VM:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 https://esm.ubuntu.com/apps/ubuntu noble-apps-security InRelease
Hit:5 https://esm.ubuntu.com/apps/ubuntu noble-apps-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:7 https://esm.ubuntu.com/infra/ubuntu noble-infra-security InRelease
Hit:8 https://esm.ubuntu.com/infra/ubuntu noble-infra-updates InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
b3ast@UltraBot-VM:~$
```

```
b3ast@UltraBot-VM:~$ sudo apt install mininet -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mininet is already the newest version (2.3.0-1.1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
b3ast@UltraBot-VM:~$
```

```
b3ast@UltraBot-VM:~$ cd ~
b3ast@UltraBot-VM:~$ git clone https://github.com/noxrepo/pox.git
Cloning into 'pox'...
remote: Enumerating objects: 13425, done.
remote: Counting objects: 100% (277/277), done.
remote: Compressing objects: 100% (50/50), done.
remote: Total 13425 (delta 258), reused 227 (delta 227), pack-reused 13148 (from 2)
Receiving objects: 100% (13425/13425), 5.14 MiB | 9.98 MiB/s, done.
Resolving deltas: 100% (8704/8704), done.
b3ast@UltraBot-VM:~$ cd pox
ls
debug-pox.py  ext  LICENSE  NOTICE  pox  pox.py  README.md  setup.cfg  tests  tools
b3ast@UltraBot-VM:~/pox$ python3 pox.py --version
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
CPython (3.12.3/Mar 3 2026 12:15:18)
```

## First time test:

```
b3ast@UltraBot-VM:~/pox$ sudo mn --test pingall
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller

*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
*** Stopping 0 controllers

*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
completed in 0.613 seconds
```

Terminal 1: POX Starting up:

```
b3ast@UltraBot-VM:~/pox$ cd ~/pox
python3 pox.py log.level --DEBUG network_monitor
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:network_monitor:Network Utilization Monitor started.
INFO:network_monitor:Waiting for switches to connect on port 6633...
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.12.3/Mar 3 2026 12:15:18)
DEBUG:core:Platform is Linux-6.17.0-20-generic-x86_64-with-glibc2.39
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
```

(and a lot of extended output, confirming that it works.)

Terminal 2:

```
*** Cleanup complete.
*** Adding controller
*** Adding switches
*** Adding hosts
*** Adding links
(10.00Mbit) (10.00Mbit) (10.00Mbit) (10.00Mbit) (10.00Mbit) (10.00Mbit) (10.00Mbit) (10.00Mbit) (20.00Mbit) *** Error: Warning: sch_htb: quantum of class 50001 is b
ig. Consider r2q change.
(20.00Mbit) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
*** Starting network
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 2 switches
s1 (10.00Mbit) (10.00Mbit) (20.00Mbit) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
s2 (10.00Mbit) (10.00Mbit) (20.00Mbit) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
...(10.00Mbit) (10.00Mbit) (20.00Mbit) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(10.00Mbit) (10.00Mbit) (20.00Mbit) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.

*** Setting OpenFlow version to 1.0
*** Waiting for switches to connect to controller...

*** Network Utilization Monitor Topology Ready ***
Hosts: h1(10.0.0.1) h2(10.0.0.2) h3(10.0.0.3) h4(10.0.0.4)
Switches: s1 <-> s2
Controller: POX at 127.0.0.1:6633

*** Running initial connectivity test
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
*** Starting CLI
*** Starting CLI:
mininet>
```

> Scenario 1

pingall showing 0% packet loss, and Single ping output showing RTT values, and POX Output:

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet> h1 ping -c 5 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.064 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.193 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.151 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.106 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.147 ms

--- 10.0.0.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4171ms
rtt min/avg/max/mdev = 0.064/0.132/0.193/0.043 ms
```

```
----- FLOW TABLE: Switch 00-00-00-00-00-01 (1 entries) -----
INFO:network_monitor:  match=ofp_match
   | actions=[<output port:65533 max_len:65535>] | packets=139 | bytes=11432 | duration=179s
INFO:network_monitor:--------------------------------------------------
INFO:network_monitor:
========== PORT STATS: Switch 00-00-00-00-00-02 ==========
INFO:network_monitor:  Port  1 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     49 | TX pkts:    120 | Drops: 0/0
INFO:network_monitor:  Port  2 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     43 | TX pkts:    114 | Drops: 0/0
INFO:network_monitor:  Port  3 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     94 | TX pkts:     92 | Drops: 0/0
INFO:network_monitor:  Active flows installed: 75
INFO:network_monitor:==================================================
INFO:network_monitor:
----- FLOW TABLE: Switch 00-00-00-00-00-02 (1 entries) -----
INFO:network_monitor:  match=ofp_match
   | actions=[<output port:65533 max_len:65535>] | packets=136 | bytes=11214 | duration=179s
INFO:network_monitor:--------------------------------------------------
INFO:network_monitor:
========== PORT STATS: Switch 00-00-00-00-00-01 ==========
INFO:network_monitor:  Port  1 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     48 | TX pkts:    121 | Drops: 0/0
INFO:network_monitor:  Port  2 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     42 | TX pkts:    115 | Drops: 0/0
INFO:network_monitor:  Port  3 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     92 | TX pkts:     94 | Drops: 0/0
INFO:network_monitor:  Active flows installed: 75
INFO:network_monitor:==================================================
INFO:network_monitor:
----- FLOW TABLE: Switch 00-00-00-00-00-01 (1 entries) -----
INFO:network_monitor:  match=ofp_match
   | actions=[<output port:65533 max_len:65535>] | packets=139 | bytes=11432 | duration=184s
INFO:network_monitor:--------------------------------------------------
INFO:network_monitor:
========== PORT STATS: Switch 00-00-00-00-00-02 ==========
INFO:network_monitor:  Port  1 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     49 | TX pkts:    120 | Drops: 0/0
INFO:network_monitor:  Port  2 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     43 | TX pkts:    114 | Drops: 0/0
INFO:network_monitor:  Port  3 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     94 | TX pkts:     92 | Drops: 0/0
INFO:network_monitor:  Active flows installed: 75
INFO:network_monitor:==================================================
INFO:network_monitor:
----- FLOW TABLE: Switch 00-00-00-00-00-02 (1 entries) -----
INFO:network_monitor:  match=ofp_match
   | actions=[<output port:65533 max_len:65535>] | packets=136 | bytes=11214 | duration=184s
INFO:network_monitor:--------------------------------------------------
```

- Scenario 2.

```
mininet> h3 iperf -s &
mininet> h1 iperf -c 10.0.0.3 -t 20
------------------------------------------------------------
Client connecting to 10.0.0.3, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.1 port 48700 connected with 10.0.0.3 port 5001 (icwnd/mss/irtt=14/1448/9115)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-21.3080 sec  23.6 MBytes  9.30 Mbits/sec
```

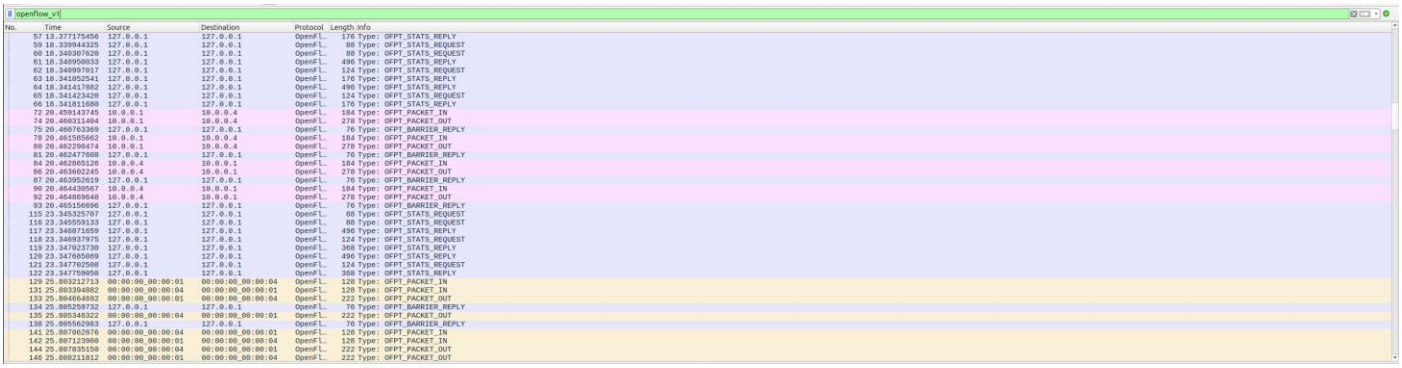
```
========== PORT STATS: Switch 00-00-00-00-00-01 ==========
INFO:network_monitor:  Port  1 | RX:  7480.20 Kbps | TX:    12.65 Kbps | RX pkts:    657 | TX pkts:    717 | Drops: 0/0
WARNING:network_monitor:  *** HIGH UTILIZATION ALERT: Switch 00-00-00-00-00-01 Port 1 RX=7.48 Mbps TX=0.01 Mbps ***
INFO:network_monitor:  Port  2 | RX:     0.00 Kbps | TX:     0.00 Kbps | RX pkts:     22 | TX pkts:     85 | Drops: 0/0
INFO:network_monitor:  Port  3 | RX:    12.65 Kbps | TX:  7480.20 Kbps | RX pkts:    688 | TX pkts:    689 | Drops: 0/0
WARNING:network_monitor:  *** HIGH UTILIZATION ALERT: Switch 00-00-00-00-00-01 Port 3 RX=0.01 Mbps TX=7.48 Mbps ***
INFO:network_monitor:  Active flows installed: 37
INFO:network_monitor:==================================================
```

## ➢ Scenario 3. dump-flow and dump-port

```
mininet> sh ovs-ofctl dump-flows s1
 cookie=0x0, duration=344.883s, table=0, n_packets=149, n_bytes=12121, priority=0 actions=CONTROLLER:65535
mininet> sh ovs-ofctl dump-flows s2
 cookie=0x0, duration=349.924s, table=0, n_packets=146, n_bytes=11903, priority=0 actions=CONTROLLER:65535
mininet> sh ovs-ofctl dump-ports s1
OFPST_PORT reply (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
  port  "s1-eth1": rx pkts=1753, bytes=17811094, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=1831, bytes=124377, drop=0, errs=0, coll=0
  port  "s1-eth2": rx pkts=43, bytes=3334, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=122, bytes=11526, drop=0, errs=0, coll=0
  port  "s1-eth3": rx pkts=1799, bytes=120945, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=1802, bytes=17815923, drop=0, errs=0, coll=0
mininet>
```

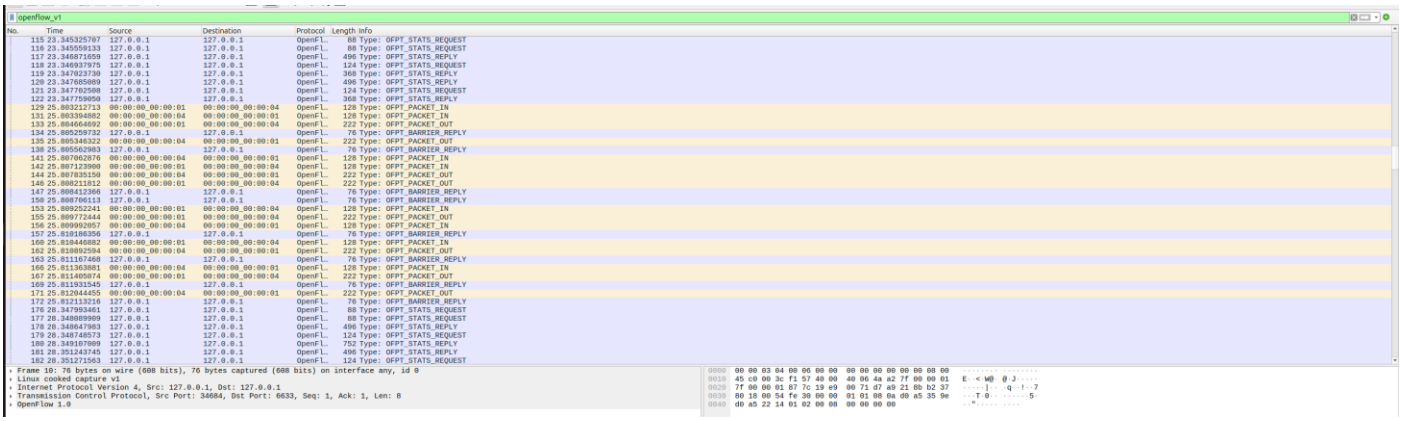
## ➢ Scenario 4.

```
mininet> h1 iperf -s &
mininet> h2 iperf -c 10.0.0.1 -t 10 &
mininet> h3 iperf -s &
------------------------------------------------------------
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.3 port 5001 connected with 10.0.0.1 port 56198 (icwnd/mss/irtt=14/1448/5268)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-15.4748 sec  16.9 MBytes  9.15 Mbits/sec
mininet> h4 iperf -c 10.0.0.3 -t 10
------------------------------------------------------------
Client connecting to 10.0.0.3, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.4 port 47408 connected with 10.0.0.3 port 5001 (icwnd/mss/irtt=14/1448/5116)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-11.0517 sec  12.4 MBytes  9.39 Mbits/sec
mininet>
```



openflow_v1

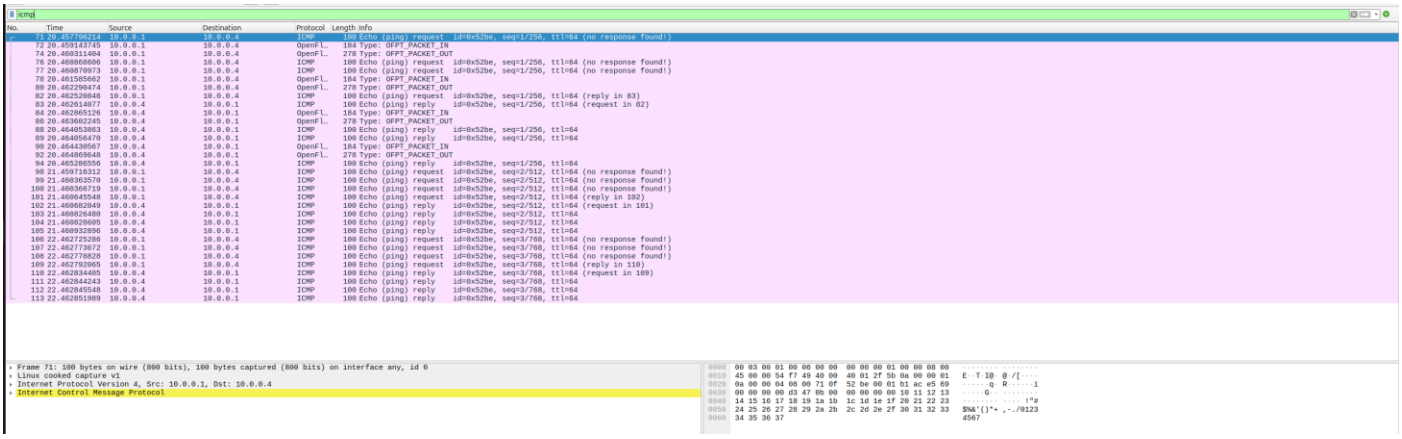
| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 57 | 13.377175456 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 176 | Type: OFPT_STATS_REPLY |
| 59 | 18.339944325 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 60 | 18.340307620 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 61 | 18.340950033 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 62 | 18.340997817 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |
| 63 | 18.341052541 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 176 | Type: OFPT_STATS_REPLY |
| 64 | 18.341417082 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 65 | 18.341423420 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |
| 66 | 18.341811680 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 176 | Type: OFPT_STATS_REPLY |
| 72 | 20.459143745 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 74 | 20.460311404 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 75 | 20.460763369 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 78 | 20.461585662 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 80 | 20.462290474 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 81 | 20.462477608 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 84 | 20.462865126 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 86 | 20.463602245 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 87 | 20.463952619 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 90 | 20.464430567 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 92 | 20.464869648 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 93 | 20.465156696 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 115 | 23.345325707 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 116 | 23.345559133 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 117 | 23.346871659 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 118 | 23.346937975 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |
| 119 | 23.347023730 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 368 | Type: OFPT_STATS_REPLY |
| 120 | 23.347685089 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 121 | 23.347702508 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |
| 122 | 23.347759050 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 368 | Type: OFPT_STATS_REPLY |
| 129 | 25.803212713 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 131 | 25.803394882 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 133 | 25.804664692 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 134 | 25.805259732 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 135 | 25.805346322 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 138 | 25.805562983 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 141 | 25.807062876 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 142 | 25.807123900 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 144 | 25.807835150 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 146 | 25.808211812 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |



openflow_v1

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 115 | 23.345325707 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 116 | 23.345559133 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 117 | 23.346871659 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 118 | 23.346937975 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |
| 119 | 23.347023730 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 368 | Type: OFPT_STATS_REPLY |
| 120 | 23.347685089 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 121 | 23.347702508 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |
| 122 | 23.347759050 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 368 | Type: OFPT_STATS_REPLY |
| 129 | 25.803212713 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 131 | 25.803394882 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 133 | 25.804664692 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 134 | 25.805259732 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 135 | 25.805346322 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 138 | 25.805562983 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 141 | 25.807062876 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 142 | 25.807123900 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 144 | 25.807835150 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 146 | 25.808211812 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 147 | 25.808412366 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 150 | 25.808706113 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 153 | 25.809252241 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 155 | 25.809772444 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 156 | 25.809992057 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 157 | 25.810186356 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 160 | 25.810446882 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 162 | 25.810892594 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 163 | 25.811167460 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 166 | 25.811363881 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 128 | Type: OFPT_PACKET_IN |
| 167 | 25.811405074 | 00:00:00_00:00:01 | 00:00:00_00:00:04 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 169 | 25.811931545 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 171 | 25.812044455 | 00:00:00_00:00:04 | 00:00:00_00:00:01 | OpenFl… | 222 | Type: OFPT_PACKET_OUT |
| 172 | 25.812113216 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 76 | Type: OFPT_BARRIER_REPLY |
| 176 | 28.347993461 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 177 | 28.348089909 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 88 | Type: OFPT_STATS_REQUEST |
| 178 | 28.348647983 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 179 | 28.348748573 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |
| 180 | 28.349107009 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 752 | Type: OFPT_STATS_REPLY |
| 181 | 28.351243745 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 496 | Type: OFPT_STATS_REPLY |
| 182 | 28.351271563 | 127.0.0.1 | 127.0.0.1 | OpenFl… | 124 | Type: OFPT_STATS_REQUEST |

▸ Frame 10: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface any, id 0
▸ Linux cooked capture v1
▸ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
▸ Transmission Control Protocol, Src Port: 34684, Dst Port: 6633, Seq: 1, Ack: 1, Len: 8
▸ OpenFlow 1.0



icmp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 71 | 20.457796314 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=1/256, ttl=64 (no response found!) |
| 72 | 20.459143745 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 74 | 20.460311404 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 76 | 20.460868606 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=1/256, ttl=64 (no response found!) |
| 77 | 20.460870973 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=1/256, ttl=64 (no response found!) |
| 78 | 20.461585662 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 80 | 20.462290474 | 10.0.0.1 | 10.0.0.4 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 82 | 20.462520846 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=1/256, ttl=64 (reply in 83) |
| 83 | 20.462614077 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=1/256, ttl=64 (request in 82) |
| 84 | 20.462865126 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 86 | 20.463602245 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 88 | 20.464053863 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=1/256, ttl=64 |
| 89 | 20.464056470 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=1/256, ttl=64 |
| 90 | 20.464430567 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 184 | Type: OFPT_PACKET_IN |
| 92 | 20.464869648 | 10.0.0.4 | 10.0.0.1 | OpenFl… | 278 | Type: OFPT_PACKET_OUT |
| 94 | 20.465286556 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=1/256, ttl=64 |
| 98 | 21.459716312 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=2/512, ttl=64 (no response found!) |
| 99 | 21.460363570 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=2/512, ttl=64 (no response found!) |
| 100 | 21.460366719 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=2/512, ttl=64 (no response found!) |
| 101 | 21.460645548 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=2/512, ttl=64 (reply in 102) |
| 102 | 21.460682049 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=2/512, ttl=64 (request in 101) |
| 103 | 21.460826400 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=2/512, ttl=64 |
| 104 | 21.460828605 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=2/512, ttl=64 |
| 105 | 21.460932896 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=2/512, ttl=64 |
| 106 | 22.462725206 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=3/768, ttl=64 (no response found!) |
| 107 | 22.462773072 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=3/768, ttl=64 (no response found!) |
| 108 | 22.462778828 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=3/768, ttl=64 (no response found!) |
| 109 | 22.462792065 | 10.0.0.1 | 10.0.0.4 | ICMP | 100 | Echo (ping) request id=0x52be, seq=3/768, ttl=64 (reply in 110) |
| 110 | 22.462834405 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=3/768, ttl=64 (request in 109) |
| 111 | 22.462844243 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=3/768, ttl=64 |
| 112 | 22.462845548 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=3/768, ttl=64 |
| 113 | 22.462851989 | 10.0.0.4 | 10.0.0.1 | ICMP | 100 | Echo (ping) reply id=0x52be, seq=3/768, ttl=64 |

▸ Frame 71: 100 bytes on wire (800 bits), 100 bytes captured (800 bits) on interface any, id 0
▸ Linux cooked capture v1
▸ Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.4
▸ Internet Control Message Protocol